Reflective Non-Exhaustion:

Certification, Realization, Fibers, and Obstruction

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Abstract

Central claim. We state and defend a summit-level claim from a machine-checked program in Lean 4: *canonical certification does not, in general, exhaust realization.* When a formal architecture has a bare certification layer, an enriched realization layer, and a sound comparison map $\pi : E \to B$, collapse at the bare carrier may coexist with nontrivial realized structure above it; the residue is organized by non-injective comparison, nontrivial fibers, sections, and obstruction laws.

Suite support. The claim is supported by six completed kinds of discharge: the internal origin theorem in Infinity Compression; external transfer across twelve benchmark families; an algebraic obstruction theorem for group extensions; an arithmetic bridge through embedding problems; a topological discharge via Quillen's Theorem A for Galois connections; and a computability anchor (halting and Rice-style barriers in `ComputablePred`).

This paper's role. It does not introduce new axioms or replace the companion formal papers. It distills the architecture into one summit statement, records what has been proved, and separates the claim from incompleteness and generic proof relevance in dependent type theory. It also explains why the collapse/refinement/fiber/obstruction pattern is a reusable research instrument.

Frontier. The next step is a strongest universal inevitability theorem over the broadest abstract class; this manuscript is synthesis of completed instances and a candidate general law, not that closure.

Suite context. The Infinity Compression paper suite is organized as follows: the *flagship manuscript* states the *internal origin theorem*; the *transferability paper* validates the same packaging on twelve external benchmarks; *general-method manuscripts* (group extensions; Quillen) supply theorem-level leverage as the *algebraic and arithmetic engine* and *topology discharge*; the *synthesis manuscript* maps the full program; the *summit manuscript* distills reflective non-exhaustion; the *technical Mathlib companion* is reviewer-facing. This manuscript is the *summit statement*.

Keywords: formal verification, certification, realization, fibers, obstruction, reflective systems, Lean 4, Mathlib

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1 The summit claim

1.1 The central statement

The program distilled in this paper is centered on the following claim.

Theorem 1.1 (Reflective Non-Exhaustion Theorem)**.** *Let a* reflective certification architecture *consist of:*

(i) a bare certification carrier B;

(ii) an enriched realization carrier E;

(iii) a sound comparison map $\pi : E \to B$;

(iv) a distinguished canonical region $C \subseteq B$ singled out by the theory's certification rules.

Then the completed formal instances studied in this program support the following uniform law—a dichotomy between total collapse along π and genuine non-exhaustion:

(a) (collapse regime) *The enriched layer carries no nontrivial residue above certification for the comparison problem governed by π.*

(b) (non-exhaustion regime) *Canonical certification collapses inside C, the realization layer remains nontrivial above that collapse, and the nontriviality is organized by non-injectivity of π, by nontrivial fibers $\pi^{-1}(b)$ for $b \in C$, and by explicit obstruction or section laws governing when collapse can or cannot be lifted.*

In the non-exhaustion regime, canonical certification does not exhaust realization. The suite's completed formal discharges instantiate this non-exhaustion regime in distinct mathematical domains.

The completed discharges justify reading this law as the governing synthesis of the current program; a strongest universal form over the broadest abstract class remains future work (section 2).
Here *reflective* means that the architecture contains both an official certification layer and a structured realization layer connected by an explicit comparison map, so that the theory can internally track the relation between what is certified and how that certification is realized.
In the non-exhaustion regime, the residue does not live on the bare carrier itself but in the fibers of π; liftability is governed by obstruction theory, not by certificate theory alone. Across the suite, this is one shared architecture, not family resemblance.
Theorem 1.1 is a summit-level synthesis statement. Its formal instances are proved in the companion papers and Lean modules cited below; this paper does not ask the reader to accept it as a slogan detached from formal mathematics.

1.2 Public paraphrase

A concise public rendering is:

> *A system's canonical certificate need not fully contain the structure by which that certificate is realized.*

A stronger paraphrase:

> *Canonical certification does not, in general, exhaust realized structure.*

1.3 What this paper is for

This paper is not the place where the detailed Lean proofs live; nor is it a popular essay. Its role is to do four things that none of the companion papers, taken alone, can do as sharply:

(1) state the permanent theorem-level idea in one place;

(2) show exactly which completed results already support it;

(3) separate it from nearby but different ideas (incompleteness, generic proof relevance, informal rhetoric);

(4) explain why the architecture is a reusable research instrument rather than a one-off observation.

1.4 Program route labels

The suite groups its formal work under named *routes* (navigation labels, not routing tables or inference rules). **Route C** is the headline reflective non-exhaustion claim; **Routes A** and **B** cover infrastructure lemmas and the algebra/topology/arithmetic general-method papers; **Route D** is the logic/computability discharge—halting and Rice scaffolding in Mathlib's `ComputablePred` layer (Lean `RouteD/`). This vocabulary is fixed here so that later mentions of Route D are not unexplained forward references.

2 Three levels of claim

This paper distinguishes three levels; readers should not conflate them.

Instance level. The companion papers prove specific theorems on named carriers in Lean.

Synthesis level. This paper identifies the common architecture those theorems instantiate and states it as the summit thesis (Theorem 1.1).

Frontier level. A stronger universal inevitability theorem over the broadest abstract class remains future work.

3 Why this counts as theorem-level synthesis

The summit claim is not a narrative imposed after the fact. It is theorem-level because:

- **Named formal instances.** Companion papers and Lean modules prove concrete theorems on named carriers (zero `sorry` in the relevant modules).
- **Shared comparison structure.** The same pattern—bare layer, enriched layer, comparison map $\pi : E \to B$—is instantiated across internal, benchmark, algebraic, arithmetic, topological, and computability-adjacent settings.
- **Fiber and obstruction law.** Residual structure is located in fibers and controlled by obstruction in each discharge, not by certificate rhetoric alone.

Theorem 1.1 is the explicit synthesis statement these instances support.

4 What is actually proved in the suite

The summit claim is supported by several completed theorem families, all formalized in Lean 4 with zero `sorry` in the relevant modules.

4.1 Internal origin theorem

The flagship Infinity Compression paper [1] proves the core certification/realization split on named carriers internal to the library:

(1) canonical collapse at the bare certification carrier;

(2) a reflective split theorem on enriched witnesses;

(3) strict refinement via a typed forgetful map;

(4) nontriviality of the canonical fiber;

(5) injective role coordinates on a standard fiber patch.

This is the *internal origin theorem*: the place where the architecture first appears not as vocabulary but as a proved, load-bearing structure.

4.2 External transfer

The validation paper [2] proves that the same packaging exports to twelve independent mathematical benchmark families: quotients, products, dependent sums, coproduct discriminants, subtypes, orbit quotients, ideal quotients, pushouts, classifiers, localization, pullbacks, and a non-surjective control case.

The role of that paper is not theorem-level domain leverage. Its role is transferability: it demonstrates that the architecture is not an idiosyncratic artifact of one flagship library.

4.3 Algebraic discharge

The group extension paper [3] proves the central algebraic obstruction law:

$$\text{splits} \iff \text{trivial cocycle}.$$

It also proves the section cocycle identity, local-to-global obstruction principles, and associated fiber structure.

This is a genuine theorem-level discharge in classical mathematics: the architecture is not merely organizational there, but mathematically load-bearing.

4.4 Arithmetic discharge

The same paper and associated Lean modules show that embedding problems are equivalent to extension splitting problems and hence to cocycle triviality:

$$\text{solvable embedding problem} \iff \text{splitting} \iff \text{trivial cocycle}.$$

This supplies the arithmetic face of the program, connecting the architecture to inverse Galois-style lifting problems.

4.5 Topological discharge

The Quillen paper [4] proves Quillen's Theorem A for Galois connections at the simplicial/nerve level, using contractible fibers and homotopy-equivalence data. This is the topological instance of the same architecture: fibers, explicit comparison maps, and a theorem-level consequence.

4.6 Computability anchor

The summit-supporting Lean modules also contain a logic/computability anchor: halting is not capturable by the relevant certification layer, and Rice-style phenomena block certification of nontrivial semantic properties in the `ComputablePred` setting. This does not mean the program is "just Gödel again," but it ensures the summit architecture touches standard non-certifiability barriers in classical computability theory.

4.7 Summary of support

Taken together, the suite already contains:

Support type	**Completed instance**
Internal origin	certification vs. realization split in IC
Transferability	twelve external benchmark families
Algebraic discharge	group extension splitting criterion
Arithmetic discharge	embedding problems and tower bridge
Topological discharge	Quillen A for Galois connections
Computability anchor	halting / Rice non-certifiability layer

That support is sufficient to justify a summit paper: the summit claim is no longer a program aspiration but a synthesis of completed formal discharges.

5 The architecture itself

5.1 Certification

A *certification layer* is a carrier of theorem-level or officially packaged invariants—the layer at which a theory certifies an object, witness, or verdict. Its defining feature here is that it may *collapse*: many legitimate routes may project to the same canonical certification object.

5.2 Realization

A *realization layer* is an enriched carrier above the certificate. It tracks the structure by which the certificate is obtained or instantiated: sections, role assignments, dependency shapes, enriched reflective witnesses, splitting data, chosen representatives, homotopy witnesses, and similar objects.

Realization is not decoration. It becomes mathematically serious exactly when the comparison map from realization to certification is not injective.

5.3 Comparison map

The *comparison map* or *forgetful map*

$$\pi : E \to B$$

forgets enriched structure and records only the bare certified image.

This map is the central organizing object. Once π is explicit, one can ask:

(a) Is π surjective?

(b) Is π injective?

(c) What do the fibers $\pi^{-1}(b)$ look like?

(d) Are there canonical sections or right inverses?

(e) What obstruction governs lifting from the bare layer back upstairs?

5.4 Fibers

The fiber over $b \in B$ is

$$\mathrm{Fib}(b) := \pi^{-1}(b).$$

When certification collapses, fibers are where residual structure lives—the usual fiber picture in topology [8]. Multiple points in $\mathrm{Fib}(b)$ mean bare certification does not fix realization uniquely.

5.5 Obstruction

An *obstruction law* is the theorem governing when one may lift from bare to enriched structure. Lifting and obstruction in the sense of fiber bundles and homotopy theory are standard [9, 8]. In different domains this obstruction appears in different mathematical clothing:

- a cocycle in group extensions,
- fiber contractibility in Quillen-style comparison,
- existence or failure of sections and right inverses in benchmark families,
- non-certifiability in computability.

The summit claim is that these phenomena instantiate one collapse/refinement/fiber/obstruction architecture, not unrelated accidents.

6 The internal origin theorem

The flagship result [1] is the place where the summit architecture is first proved, rather than merely observed.

6.1 What happens internally

At the bare carrier, the standard Phase-2 extraction collapses to a unique canonical certificate. At the enriched carrier, distinct reflective witnesses remain distinct. A typed forgetful map from enriched reflective splits to canonical bare certificates is sound but not injective. The canonical fiber above the bare basepoint is nontrivial, and role data injects along a standard subfamily.

6.2 Why that matters

This gives the program its sharpest internal lesson:

> Once collapse at the bare carrier is proved, comparison questions that still matter must move upstairs to enriched carriers and fibers.

That is a mathematical, not stylistic, conclusion. The internal origin theorem is what turns the rest of the suite into a coherent program rather than a collection of analogies.

7 Transferability

The benchmark survey [2] proves that the architecture exports to many ordinary mathematical constructions.

7.1 Why transfer matters

Without transfer, the flagship theorem could be dismissed as an artifact of an idiosyncratic internal formalism. The tranche survey blocks that dismissal. It shows that the same language of:

$$\text{bare layer} \quad \text{vs.} \quad \text{enriched layer} \quad \text{vs.} \quad \text{fiber / section / right inverse}$$

is not local to one formal development.

7.2 What transfer does not claim

Transferability is not itself the summit theorem. Nor does it prove new deep classical results in each benchmark family. Its function is narrower and essential: it validates the architecture as a reusable interface.

8 Algebraic and arithmetic discharge

8.1 Group extensions

The group extension theorem

$$\text{splits} \iff \text{trivial cocycle}$$

is a model example of the summit architecture:

- the bare layer is set-theoretic section existence, which always holds;
- the enriched layer is homomorphic splitting;
- the comparison is the passage from structured splitting data to bare section data;
- the obstruction is the section 2-cocycle.

This is exactly the sort of result that makes the architecture mathematically load-bearing rather than merely descriptive.

8.2 Embedding problems

Embedding problems sharpen the picture:

$$\text{solvable} \iff \text{splits} \iff \text{trivial cocycle.}$$

Here the same architecture governs an arithmetic lifting problem. Bare data does not decide liftability by itself; the residue sits in the obstruction.

8.3 Arithmetic significance

This is important for the summit claim because it shows that the architecture has teeth in a domain that is not just internal formal theory and not just generic category packaging. It reaches into a recognized arithmetic problem class.

9 Topological discharge

The Quillen paper [4] provides the topological discharge.

9.1 What happens there

For Galois connections, the relevant fibers are principal down-sets with terminal or maximal structure, hence contractible nerves. This gives the comparison theorem: the induced nerve maps carry explicit homotopy-equivalence data.

9.2 Architectural reading

In that setting:

- the comparison map is the induced nerve map;
- the fibers encode the local comparison geometry;
- contractibility is the obstruction-theoretic condition that trivializes the comparison problem.

This is a very different mathematical domain from group extensions, yet the same architecture reappears.

10 Computability anchor

Route D (section 1.4) is the computability discharge: halting and Rice-style facts in Mathlib's `ComputablePred` layer, formalized in `RouteD/`. They show semantic phenomena that need not be fully capturable by the relevant certification apparatus, with classical landmarks in incompleteness and undecidability [5, 6, 7]. This prevents the summit architecture from being mistaken for a vague philosophical analogue of Gödel's theorem [5].

In parallel with the other domains, the bare layer is the certifiable predicate interface; the enriched layer is semantic behavior of computations; the comparison is certification through `ComputablePred`; the obstruction is classical undecidability (halting / Rice).

10.1 What is not claimed

The summit theorem is not Gödel's incompleteness theorem [5], nor is the present program claiming to subsume all of classical recursion theory. The anchor is narrower: the suite's language touches standard impossibility terrain in an honest, formal way.

11 Why this is not just Gödel

This distinction is central.

11.1 Gödel's contribution

Gödel shows that sufficiently rich formal systems face unavoidable limits: undecidable statements, incompleteness, and non-recursive truth phenomena [5].

11.2 This program's contribution

The present program shows something structurally different:

(1) where collapse occurs;

(2) where it provably does not occur;

(3) how the non-collapse is organized.

That is a positive architecture theorem, not merely a barrier theorem.

11.3 The sharp difference

Gödel's theorem says, roughly, that certain truths outrun formal provability. The summit architecture addresses a different question: where certification collapses at one carrier, realized structure may still be nontrivial upstairs, with that nontriviality organized by fibers and obstructions (Theorem 1.1, non-exhaustion regime).

The two ideas are related in spirit once **Route D** (halting and Rice in `ComputablePred`) is in play, but they are not the same theorem.

12 Why this is not merely proof relevance in type theory

A common dismissal would be: of course enriched objects exist in dependent type theory; proof relevance is built in.

That is not the result.

12.1 What would be trivial

It is trivial that richer carriers can be defined in DTT. It is trivial that one can invent a forgetful map from a rich type to a poor one.

12.2 What is nontrivial here

What is nontrivial is that the library proves:

(1) collapse at one named carrier;

(2) non-collapse at another named carrier;

(3) non-injectivity of the explicit comparison map;

(4) nontriviality and partial classification of the corresponding fibers;

(5) parallel obstruction laws in external mathematics.

The summit claim is therefore not "proof relevance exists." It is:

> *this is the correct theorem-level architecture for a broad and growing class of comparison problems.*

13 Why this changes mathematical practice

The strongest reason to take the summit claim seriously is methodological.

13.1 The reusable recipe

The architecture gives a reusable research workflow:

(1) identify the bare certificate carrier B;

(2) identify an enriched carrier E;

(3) define the comparison map $\pi : E \to B$;

(4) prove whether collapse occurs at B;

(5) test injectivity or non-injectivity of π;

(6) if non-injective, study the fibers $\mathrm{Fib}(b)$;

(7) identify sections, right inverses, or obstruction laws governing liftability.

13.2 Why this matters

This turns vague "more structure upstairs" claims into a precise program: it tells researchers what to formalize and where. The architecture is therefore a *research instrument*—a stable way to formulate and attack comparison problems where theorem-level invariants do not exhaust realizing structure.

14 Cross-domain dictionary

The following table is a *heuristic* index: it is not a formal theorem, only a compact navigation aid.

Domain	**Bare layer**	**Enriched layer**	**Comparison map**	**Fiber / obstruction**
IC flagship	canonical bare certification	enriched reflective split	typed forgetful map π_A	nontrivial canonical fiber
Benchmarks	baseline representability	structured witness bundle	projection / forgetting	fibers, sections, right inverses
Group extensions	set-theoretic section layer	splitting data	forgetting homomorphic structure	cocycle obstruction
Embedding problems	bare target quotient data	lifting / solution data	solution-to-extension comparison	solvable iff trivial cocycle
Quillen GC	order-complex / nerve image	homotopy witness data	induced nerve maps	contractible fibers
Computability	certifiable predicate layer	realized semantic behavior	certification comparison	halting / Rice barriers

It is nonetheless the best concise representation of why the summit architecture is plausible as a unifying language.

15 Exact scope

The summit claim is strong, but it is not unlimited. The following paragraphs make the *synthesis*-level claim of section 2 exact and separate it from the *frontier* level.

15.1 What is claimed

We claim that:

(1) the collapse/refinement/fiber/obstruction architecture is formally realized in the completed companion work;

(2) it transfers across multiple external families;

(3) it yields theorem-level leverage in algebra, arithmetic, topology, and computability-adjacent settings;

(4) it is therefore a plausible permanent language for a broad class of comparison problems.

15.2 What is not claimed

We do *not* claim:

(1) a single current Lean theorem quantifying over all domains at once;

(2) that every formal system must instantiate this exact architecture;

(3) that the current program has already achieved universal adoption or full Gödel-level social impact.

Outside scope. Possible speculative applications to self-modeling or phenomenological systems lie outside the formal scope of this paper; they are not part of the proved theorem content.

15.3 Why this is enough

These non-claims do not weaken the summit paper's purpose. A summit statement should mark the strongest theorem-level language already earned by the completed program, not pretend to have finished all mathematically natural extensions.

16 Open frontier

The next frontier is clear.

16.1 Mathematical extensions

The strongest natural next steps are:

(1) fuller universal quantification over a broad abstract architecture class;

(2) stronger fiber-classification theorems;

(3) completion of the H^1/H^2 bridge in Mathlib;

(4) deeper arithmetic realization through concrete Galois tower infrastructure;

(5) general Quillen Theorem A beyond the Galois-connection case.

16.2 Program-level extensions

At the level of program closure, the remaining work is not to invent an entirely new idea, but to strengthen and universalize what is already visible:

- sharper inevitability theorems,
- stronger unified statements,
- broader accepted examples,
- publication and adoption.

17 Conclusion

The program's permanent theorem-level claim is now stated in one place: Theorem 1.1.

It is not a slogan about proof objects, a rebranding of Gödelian incompleteness, or an artifact of one library. Six completed discharges support it (internal origin, transferability, algebra, arithmetic, topology, computability).

> *Collapse downstairs, structured comparison upstairs, and obstruction in the fibers is a real, reusable theorem-level architecture.*

What survives collapse has a precise mathematical home in the fibers of comparison, governed by obstruction. The right language is certification together with the geometry of what certification forgets.

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